Unit 1 Biology AS

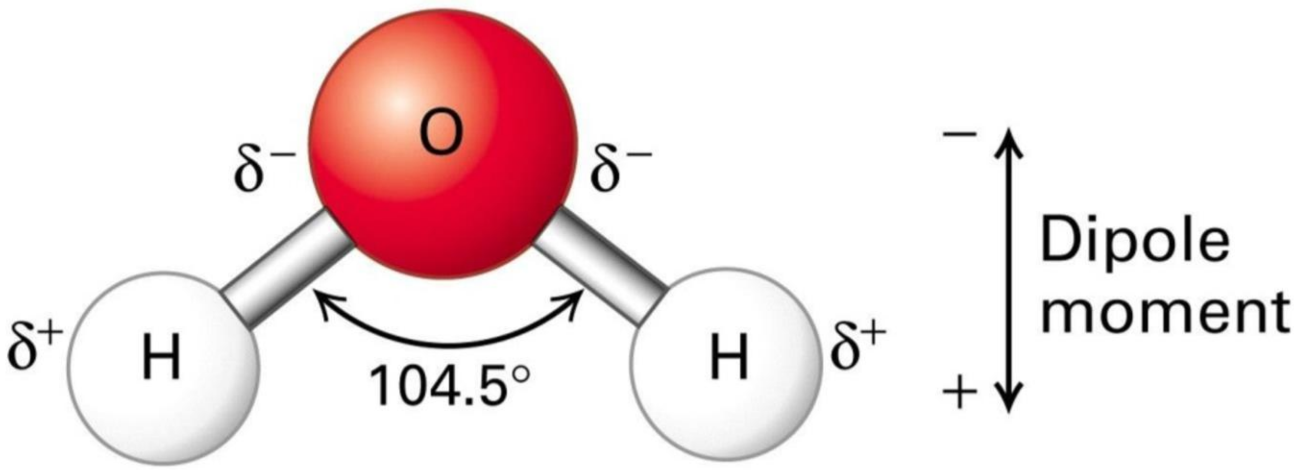
Water

Water is a polar molecule with a special type of intermolecular force known as hydrogen bonding. Water molecules form hydrogen bonds which gives it its unique characteristics.

1.1 Importance of Water as a Solvent in Transport

- **Dipole nature**: Water (H_2O) is a polar molecule — oxygen is more electronegative than hydrogen, creating a partial negative charge on oxygen and a partial positive charge on hydrogen.
- **Hydrogen bonding**: Polarity allows hydrogen bonds to form between water molecules.
- **Solvent property**: Ionic compounds (e.g. NaCl) and polar molecules (e.g. glucose) dissolve in water because water molecules surround and separate the solute ions/molecules
- **Transport**: In blood and xylem, water dissolves:
    - Gases like O_2 and CO_2
    - Ions like Na^+, Cl^-, K^+
    - Nutrients like amino acids, glucose
- **Thermal properties**: Water has a high specific heat capacity due to the presence of hydrogen bonds between its molecules. These hydrogen bonds are the strongest type of intermolecular force, requiring a significant amount of energy to break. As a result, water can absorb and store large amounts of heat without experiencing rapid temperature changes. This property plays a vital role in stabilizing the internal temperatures of living organisms and maintaining stable environmental conditions, particularly in aquatic ecosystems. Additionally, water exhibits a unique density behavior: it reaches its maximum density at 4°C. As it cools further and freezes, the hydrogen bonds arrange water molecules into a crystalline structure that is less dense than liquid water, causing ice to float. This insulating layer of ice on the surface of bodies of water helps protect aquatic life from freezing temperatures during winter.
- **Cohesion/adhesion**: Water has the ability to **stick to itself** and to **other surfaces** due to hydrogen bonding. These properties are known as **cohesion** and **adhesion**, and they play a critical role in the transport of water in plants.
- **Cohesion** refers to the attraction between water molecules. Because of hydrogen bonds, water molecules are "pulled" together, forming a continuous column of water.
- **Adhesion** refers to the attraction between water molecules and the surface, for example: walls of the xylem vessels in plants.

  In the **xylem**, these two forces work together to enable **capillary action** — the upward movement of water from the roots to the leaves. As water evaporates from the leaf surface (transpiration), cohesive forces pull more water molecules upward, while adhesive forces help water stick to the xylem walls, preventing the column from breaking.



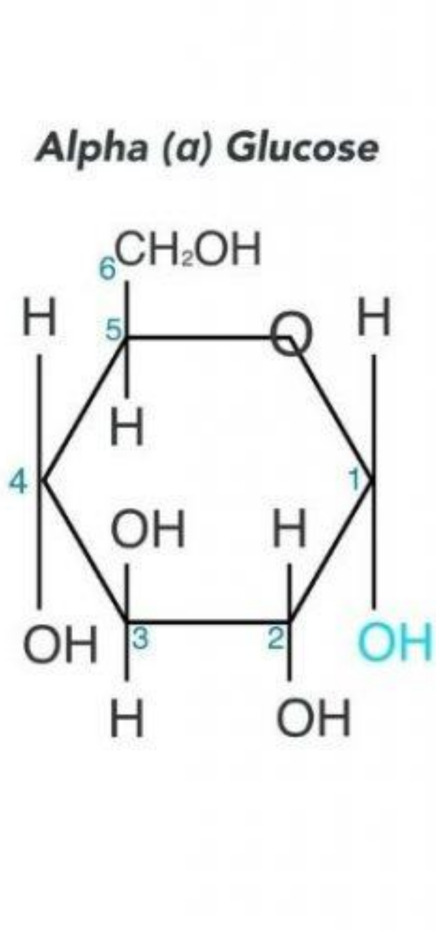
1.2 Carbohydrates: Types and Structure-Function Relationship

(i) Types:

- **Monosaccharides**: Single sugar units — e.g. glucose, fructose, galactose ($C_6H_{12}O_6$).
- **Disaccharides**: Two monosaccharides — e.g. maltose (glucose + glucose), sucrose (glucose + fructose), lactose (glucose + galactose).
- **Polysaccharides**: Many monosaccharides — e.g. glycogen (animals), starch (plants – includes amylose and amylopectin).

(ii) Structure-Function:

- **Glucose**: Small, soluble — easily transported and metabolised for energy.
- **Starch**:
  - *Amylose*: Unbranched, helical — compact for storage.
  - *Amylopectin*: Branched — rapid hydrolysis by enzymes.
- **Glycogen**: Highly branched — allows rapid glucose release in animals.
- All use **α-glucose**.
- *Note*: β-glucose and cellulose are **not required** for Topic 1.



Amylose vs amylopectin

- Both are polysaccharides of glucose
- As glucose can be broken down in aerobic respiration to produce ATP and release energy, both store energy
- Amylose is unbranched and linear because it has only 1,4-glycosidic bonds, so it is hydrolysed slower and releases energy more slowly over time
- However, amylopectin is branched due to the presence of 1,6-glycosidic bonds. So, it is compact and can be stored within a small space (inside a cell)
- This also allows quicker energy release because enzymes can hydrolyse the glycosidic bonds faster, releasing glucose more quickly for aerobic respiration
- Both are insoluble in water, so they do not affect the osmotic balance of cells

Glycogen vs starch

Feature	Glycogen	Starch
Organism	Found in animals (especially liver and muscles)	Found in plants (storage in roots, seeds, etc.)
Function	Short-term energy storage in animals	Long-term energy storage in plants
Monomer Unit	Composed entirely of α-glucose	Composed entirely of α-glucose
Glycosidic Bonds	Contains 1,4 and 1,6 glycosidic bonds	Same: both amylose and amylopectin have 1,4 bonds, amylopectin also has 1,6 bonds
Branching	Highly branched (more frequent 1,6 bonds every 8-10 glucose units)	Moderately branched (amylopectin has branches every 20-30 units); amylose is unbranched
Structure	Compact, spherical due to extensive branching	Amylose: helical; Amylopectin: branched, but less than glycogen
Solubility	Insoluble	Insoluble
Enzyme Access	Rapid hydrolysis due to many branch ends (more sites for enzyme action) → quick glucose release	Amylopectin is slower to hydrolyse; amylose is slowest due to compact helical structure
Storage Adaptation	Supports high metabolic rate of animals needing rapid energy bursts (e.g., muscle contraction)	Ideal for slow-release energy in plants during respiration

1.3 Core practical 1

1.4 Formation and Breakdown of Carbohydrates

- **Condensation reaction**: Forms glycosidic bonds between monosaccharides; water is removed.
  - E.g. glucose + glucose → maltose + H_2O
- **Hydrolysis**: Breaks glycosidic bonds using water.
  - Polysaccharides → disaccharides → monosaccharides

Bond	Example
1,4-glycosidic	Amylose
1,6-glycosidic	Amylopectin, glycogen

1.5 Lipids – Triglycerides and Saturation

(i) Triglyceride Formation:

- 1 glycerol + 3 fatty acids → **triglyceride** + 3 H_2O
- Linked by **ester bonds** formed via condensation reactions.

(ii) Saturated vs Unsaturated:

- **Saturated**: No double bonds between carbons; solid at room temp (e.g. butter).
- **Unsaturated**: One or more C=C double bonds; liquid at room temp (e.g. oils).
- Health link: Saturated fats raise LDL cholesterol → CVD risk.
- Saturated fatty acids are linear, while unsaturated fatty acids are kinked
- Saturated fatty acids have a lower Carbon to Hydrogen ratio

Ways in which fatty acids can differ from each other

- ❖ Their carbon-carbon chain lengths can differ by number of carbon atoms.
- ❖ Can be saturated or unsaturated
- ❖ Number of carbon-carbon double bonds(C=C) between unsaturated fatty acids can differ
- ❖ Linear or kinked (depending on saturated and unsaturated)

Define mass transport:

The mass flow of all components of blood, such as glucose, oxygen, Red blood cells, amino acids, platelets etc through the vessels to and from the heart

1.6 Why Animals Need a Heart and Circulation (Mass Transport)

- **Diffusion** is insufficient in large animals due to:
  - Small surface area to volume ratio
  - High metabolic demand
  - Distance between gut and cells is too high (high diffusion distance causes slow diffusion)
  - Distance between cells and skin surface is too high, skin surface is too thick to allow efficient and fast diffusion of gases across
- **Mass transport** system required to:
  - Move O_2, nutrients to cells rapidly

- Remove CO_2 and waste

- **Heart**: Muscular pump that creates pressure to circulate blood efficiently.
- Blood vessels form a closed circulatory system.

1.7 Structure and Function of Blood Vessels

There are three major types of blood vessels: arteries, veins, and capillaries — each adapted to its specific function.

Arteries

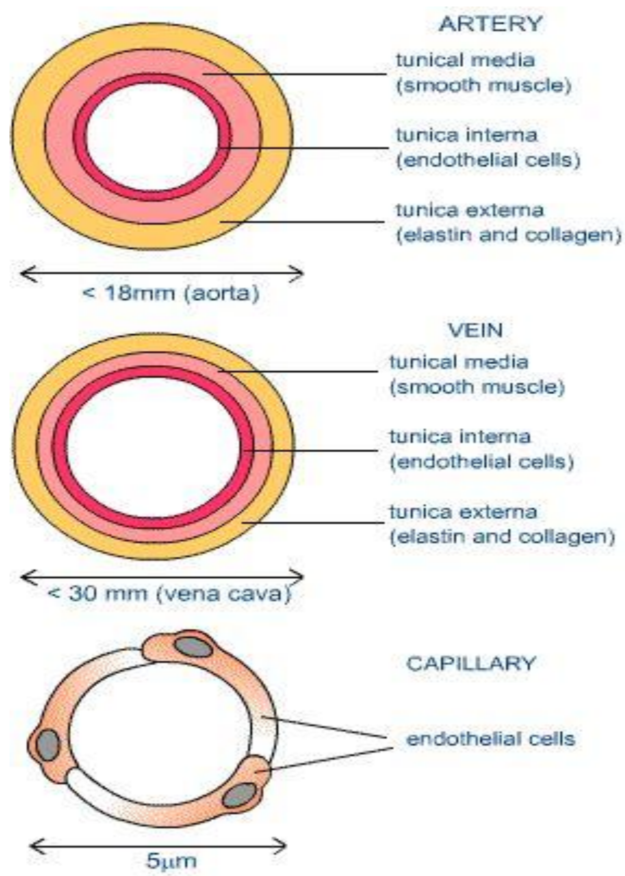
- Carry blood **away** from the heart, typically oxygenated (except pulmonary artery).
- **Structure**:
  - **Thick muscular walls** and **elastic tissue**: withstand and smooth out high pressure from the heartbeat.
  - **Elastin in the walls:** Expand due to force and pressure from the heart, recoil to exert and maintain high blood pressure, expansion helps to withstand high blood pressure and prevents rupture of artery
  - **Muscle tissue in walls**: Adjust diameter of lumen by contracting and relaxing. Contracting reduces lumen diameter, reducing blood flow to a particular tissue, therefore allowing to divert blood flow and helps in vasoconstriction. Relaxation of the muscles increases lumen diameter, allowing a greater volume of blood to flow. This helps withstand high blood pressure and high-volume blood flow, as well as increase the flow of blood to a particular tissue where rate of metabolic reactions such as respiration is higher (for example, muscle tissue and cells during exercise). So more oxygen and glucose is delivered to the cells
  - **Narrow lumen**: maintains high pressure.
  - **Endothelium**: smooth, reduces friction.
- **Function**: Rapid, high-pressure delivery of blood to organs.

Veins

- Carry blood **towards** the heart, typically deoxygenated (except pulmonary vein).
- **Structure**:
  - **Thin muscular and elastic layers**: low-pressure environment.
  - **Wider lumen**: reduces resistance to flow.
  - **Valves**: prevent backflow due to low pressure.
- **Function**: Return blood to the heart, often aided by skeletal muscle contractions.

Capillaries

- Tiny vessels connecting arteries to veins. One cell thick
- **Structure**:
  - **Single layer of endothelium**: allows efficient gas/nutrient/waste exchange by reducing diffusion
  - **Extensive network**: increases surface area.
  - **Porous:** allows leakage of tissue fluid and plasma to the surrounding tissues
- **Function**: Site of exchange between blood and tissues.



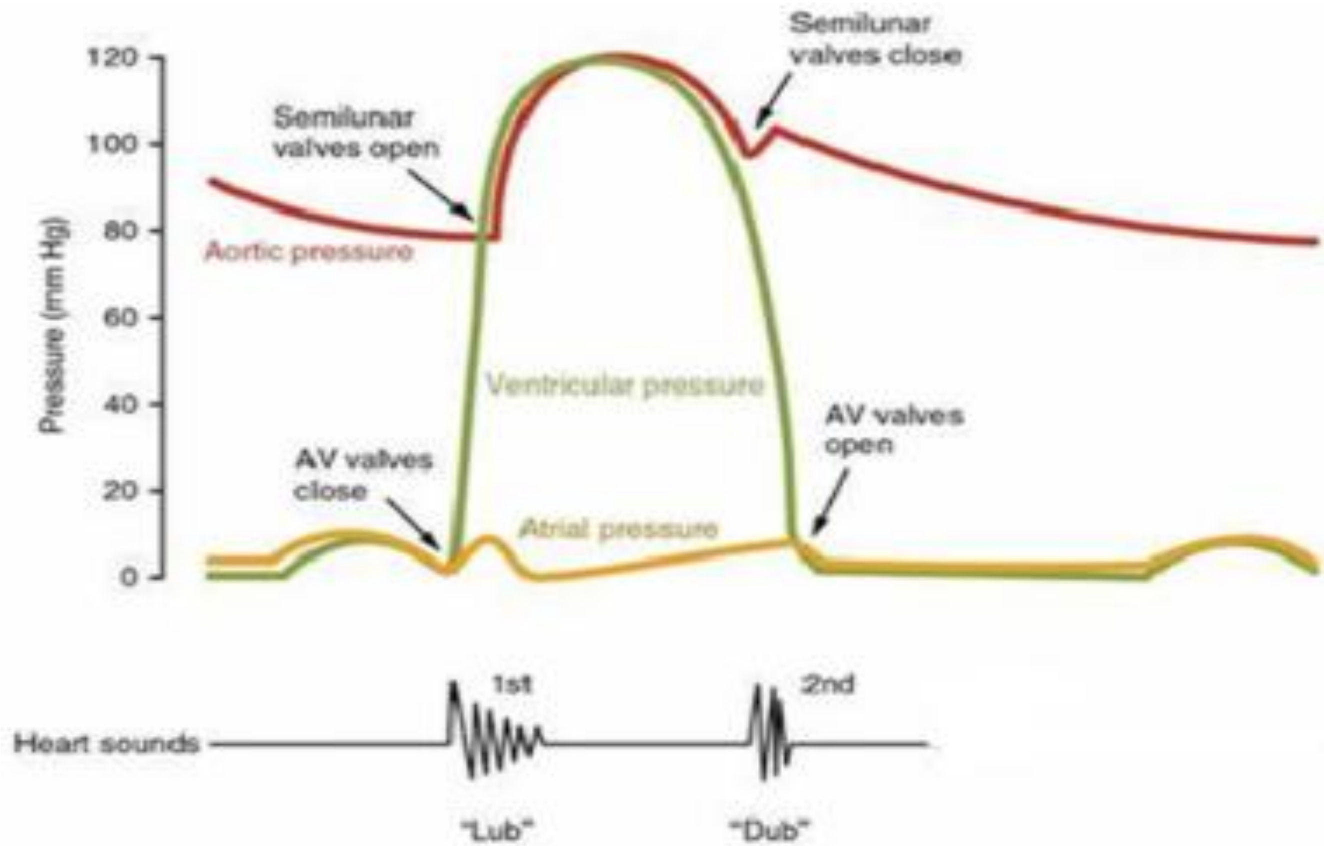
1.8 The Cardiac Cycle and Heart Function

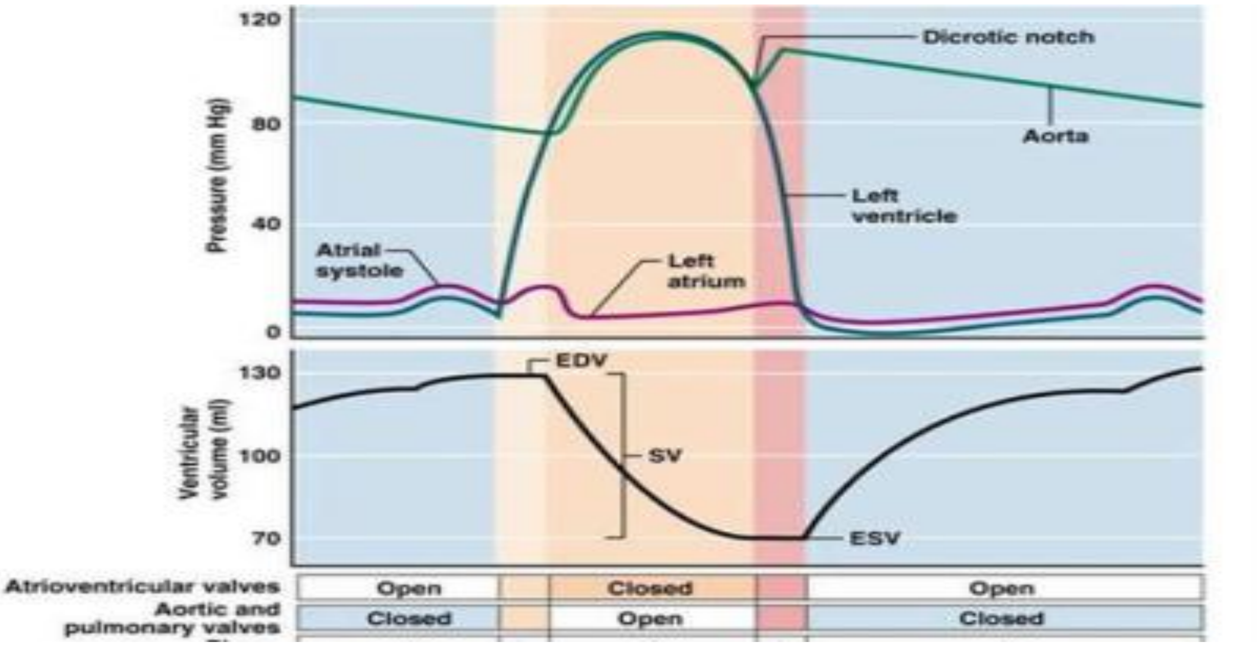
The heart operates via a cyclic sequence of contraction and relaxation phases:

Phases of the Cardiac Cycle:

1. **Atrial Systole**:
   - Atria contract.
   - Blood pushed through **atrioventricular (AV) valves** into ventricles.
   - Ventricles are relaxed.

2. **Ventricular Systole**:
    - Ventricles contract.
    - AV valves shut to prevent backflow (producing "lub" sound).
    - **Semilunar valves** (in aorta and pulmonary artery) open.
3. **Diastole**:
    - All chambers relax.
    - Blood flows passively into atria and then into ventricles as AV valves open.
    - Semilunar valves close (producing "dub" sound).





(Ignore EDV SV and ESV)

Pressure and volume changes in the heart during cardiac cycle

1. Atrial Systole (approx. 0.1 seconds)

What happens:

- **Atria contract.**
- **Blood is pushed into relaxed ventricles.**
- **AV (atrioventricular) valves are open.**
- **Semilunar valves (in aorta/pulmonary artery) are closed.**

Pressure/Volume changes:

Chamber/Vessel	Pressure Change	Volume Change	Explanation
Atria	Slight increase	Slight decrease	Muscle contraction raises pressure, forcing blood into ventricles
Ventricles	Slight increase	Increase	Blood enters from atria (passive + active filling)
Aorta & Pulmonary Artery	No change	No change	Semilunar valves closed, so no flow yet

2. Ventricular Systole (approx. 0.3 seconds)

What happens:

- **Ventricles contract.**
- **AV valves close (prevent backflow) — creates "lub" sound.**
- **When ventricular pressure > aortic/pulmonary pressure, semilunar valves open and blood is ejected.**

Pressure/Volume changes:

Chamber/Vessel	Pressure Change	Volume Change	Explanation
Ventricles	Rapid increase	Rapid decrease	Powerful contraction increases pressure → blood ejected into arteries
Atria	Slight increase	Slight increase	Atria begin to fill passively from veins (vena cava, pulmonary vein)
Aorta & Pulmonary Artery	Increase	Increase then decrease	Receive blood from ventricles → pressure rises sharply and then slowly drops as blood moves onward

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3. Diastole (approx. 0.4 seconds)

What happens:

- **All chambers relax.**
- **Ventricular pressure drops below arterial pressure → semilunar valves close → "dub" sound.**
- **Ventricular pressure drops below atrial pressure → AV valves open, ventricles fill passively.**

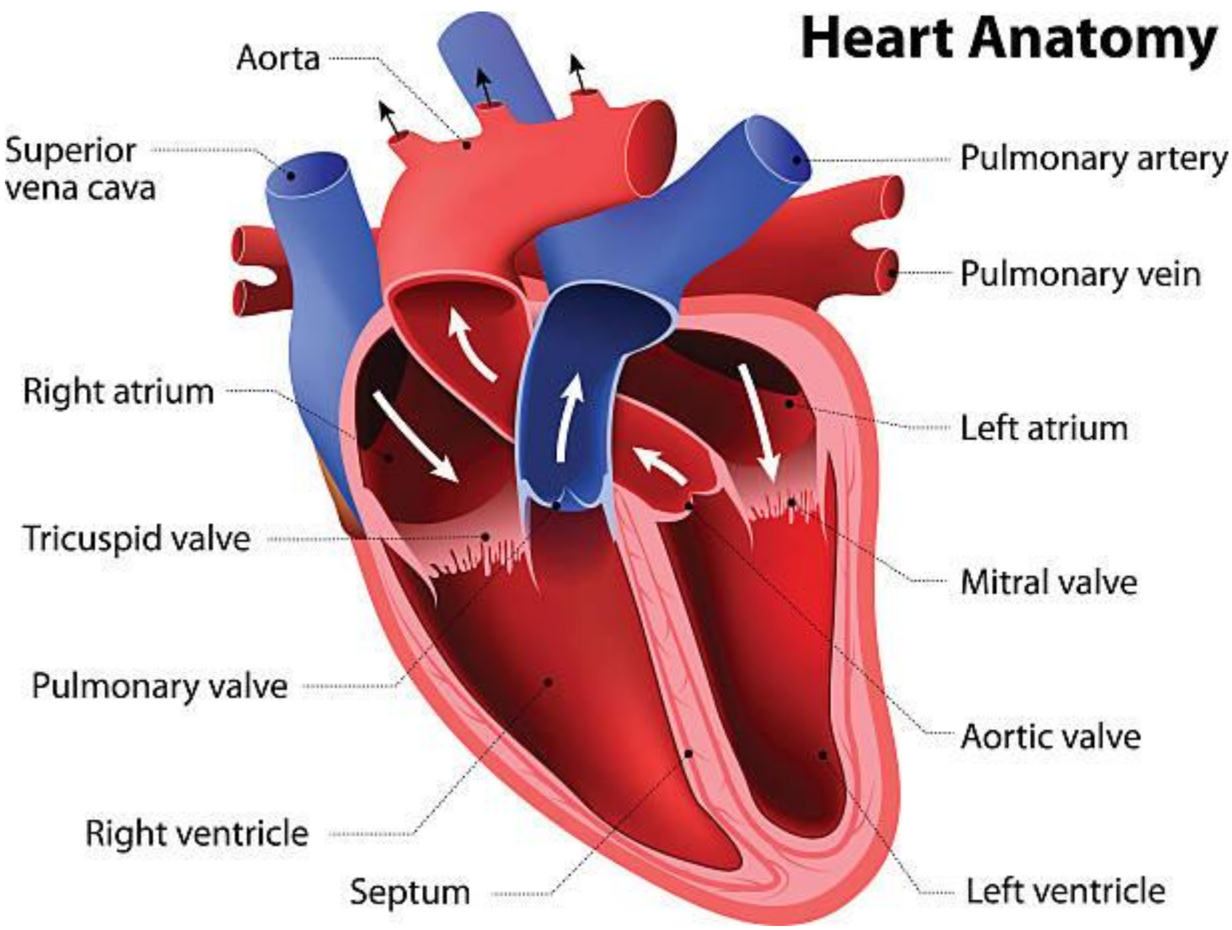
Pressure/Volume changes:

Chamber/Vessel	Pressure Change	Volume Change	Explanation
Ventricles	Decrease	Increase	Relaxed ventricles fill passively from atria
Atria	Decrease	Increase	Blood from veins continues to fill atria
Aorta & Pulmonary Artery	Decrease	Decrease	Blood moves away through circulation, pressure falls as semilunar valves close

Structure of the Heart:

- Four chambers: right and left atria (upper), right and left ventricles (lower).
- Divided into right and left side by a **septum**
- Major vessels:
    - **Aorta**: oxygenated blood to the body.
    - **Pulmonary artery**: deoxygenated blood to lungs.
    - **Pulmonary vein**: oxygenated blood from lungs.
    - **Vena cava**: deoxygenated blood from the body.
- Contains **valves** which prevents **backflow** of blood

Note: Myogenic stimulation (the ability of heart muscle to contract without nervous input) is **not required** at IAS level.



Note: pulmonary valve or aortic valve is semi-lunar valve

1.9 Role of Haemoglobin in Oxygen and Carbon Dioxide Transport

Haemoglobin (Hb):

- A **globular protein** with four polypeptide chains, each containing a **haem group** (iron-containing), which binds oxygen.
- Found in red blood cells.
- Polypeptide chains are joined to each other by disulfude bonds
- A greater proportion of amino acids with hydrophillic R groups are located on the outer region, allowing haeomoglobin to dissolve in water. A greater proportion of amino acids with hydrophobic R groups are located on the inside, due to hydrophobic (water-repelling) interactions between them (causing them to clump up on the inside, away from the external aqueous environment)

Oxygen Transport:

- **High pO_2 (partial pressure of O_2)** (e.g., lungs): haemoglobin binds O_2 → oxyhaemoglobin.
- **Low pO_2** (e.g., tissues): haemoglobin releases O_2 for respiration.

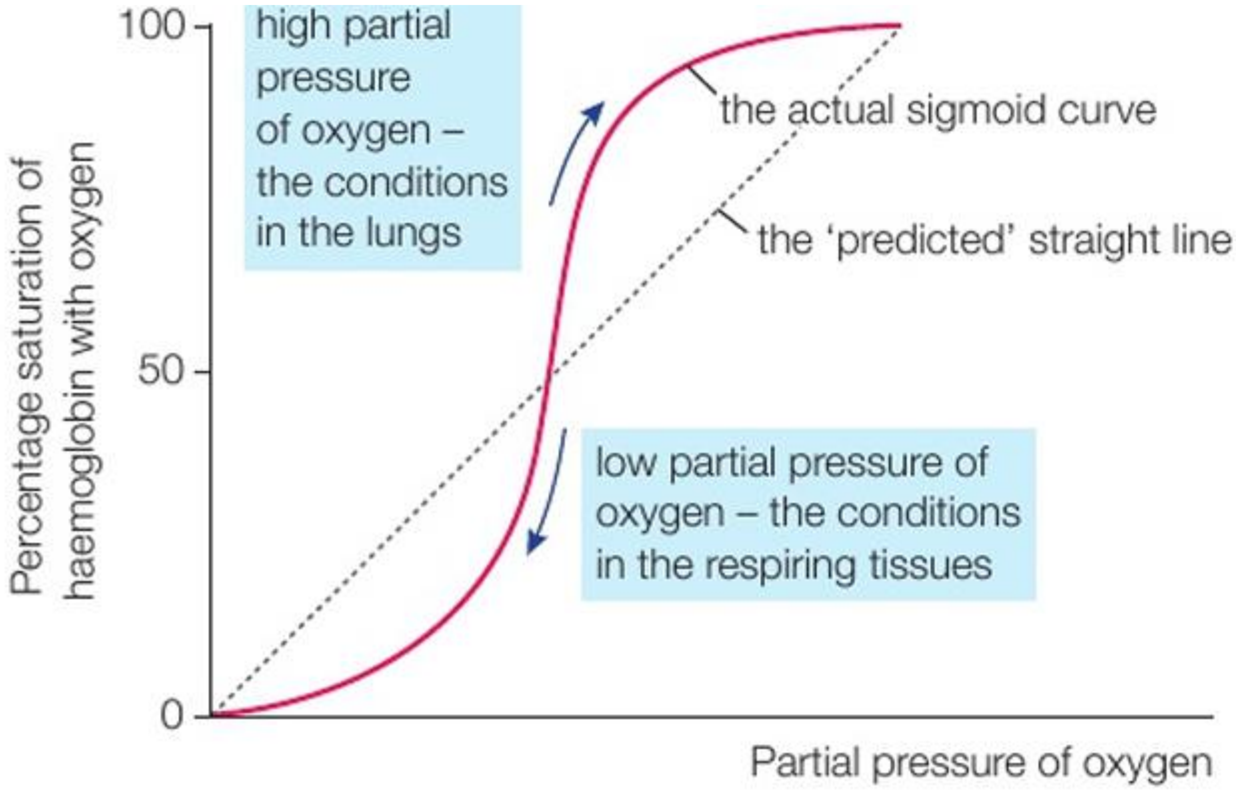
Carbon Dioxide Transport:

- Transported as:
  - **Carbonic acid/bicarbonate ions** (main method)
  - Bound to haemoglobin as **carbaminohaemoglobin**
  - Dissolved in plasma

1.10 Oxygen Dissociation Curve, Bohr Effect, Fetal vs Adult Hb

Oxygen Dissociation Curve:

- S-shaped (sigmoidal) due to cooperative binding: binding of first O_2 makes further binding easier.
- Shows % saturation of Hb with O_2 at different pO_2 levels.



Sigmoid (S-shaped) Nature of the Curve

1. **Low pO_2 – Initial shallow slope**:
   - At low oxygen tension (e.g., in **resting tissues**, $pO_2 \approx 20$–40 mmHg), **haemoglobin has a low affinity** for O_2.
   - The **first oxygen molecule binds with difficulty** because the haem group is **partially enclosed** within the globin structure.
   - Binding of the first O_2 **leads to conformational changes** of the haemoglobin molecule.
2. **Medium pO_2 – Steep middle region**:
   - Conformational change from first O_2 binding **exposes other haem groups**, making it **easier for the second and third O_2 molecules to bind**.
   - This is called **cooperative binding**.
   - Result: a **steep increase in % saturation** for a small rise in pO_2.
   - This region corresponds to **actively respiring tissues** where oxygen is readily unloaded due to slight changes in pO_2.
3. **High pO_2 – Plateau at top of curve**:
   - At high pO_2 (e.g., in **alveoli of lungs**, $pO_2 \approx 100$ mmHg), haemoglobin becomes **almost fully saturated** (close to 100%).
   - Binding of the **fourth oxygen** is more difficult because:

- Fewer available binding sites remain.
- The saturation approaches a **maximum limit**.

Bohr Effect:

- In high CO_2 areas (e.g., muscles), Hb releases O_2 more readily.
- Curve shifts **right** — decreased affinity for O_2.
- Advantageous during exercise and respiration because more carbon dioxide is released as a bi product of respiration (at high rates)

Fetal vs Adult Haemoglobin:

- **Fetal haemoglobin** has a **higher affinity** for O_2 than adult Hb.
- Enables efficient O_2 transfer from mother's blood across the placenta.
- At the same partial pressure of oxygen, affinity of fetal haemoglobin to oxygen is greater than that of the mother's adult haemoglobin
- So, oxygen is released from adult haemoglobin and binds to fetal haemoglobin, so that the fetus receives oxygen for aerobic respiration

Shifts in the Oxygen Dissociation Curve of Haemoglobin

Increased CO_2 Concentration → Curve Shifts to the Right

Mechanism (Bohr Effect):

- In respiring tissues, **CO_2 concentration is high**.
- CO_2 diffuses into red blood cells and reacts with water:

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$$

- The release of **H^+ ions lowers pH** (acidifies the environment).
- Both **H^+ ions and CO_2 bind to haemoglobin**, causing a **conformational change** that **reduces its affinity for oxygen**.

Result:

- Haemoglobin **releases oxygen more readily**.
- Curve shifts **right**, meaning **for the same pO_2, haemoglobin is less saturated**.
- This ensures **more oxygen is delivered to actively respiring tissues**.

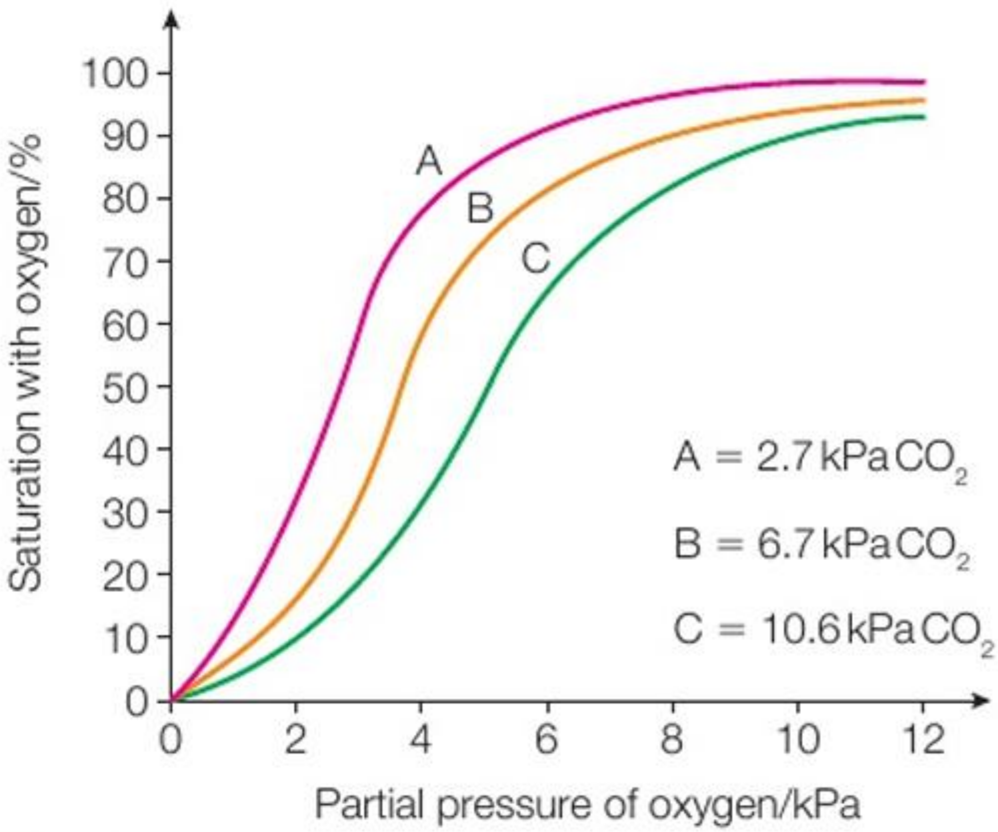
Decreased CO_2 Concentration → Curve Shifts to the Left

Mechanism:

- In the **lungs**, CO_2 is exhaled → **low CO_2 concentration** in pulmonary capillaries.
- Less carbonic acid formed → **higher pH**.
- Haemoglobin is in a state that **favors oxygen binding**.

Result:

- Haemoglobin has a **higher affinity** for oxygen.
- Curve shifts **left**, meaning **more oxygen binds at the same pO_2**.
- Enhances **oxygen uptake in the lungs**.

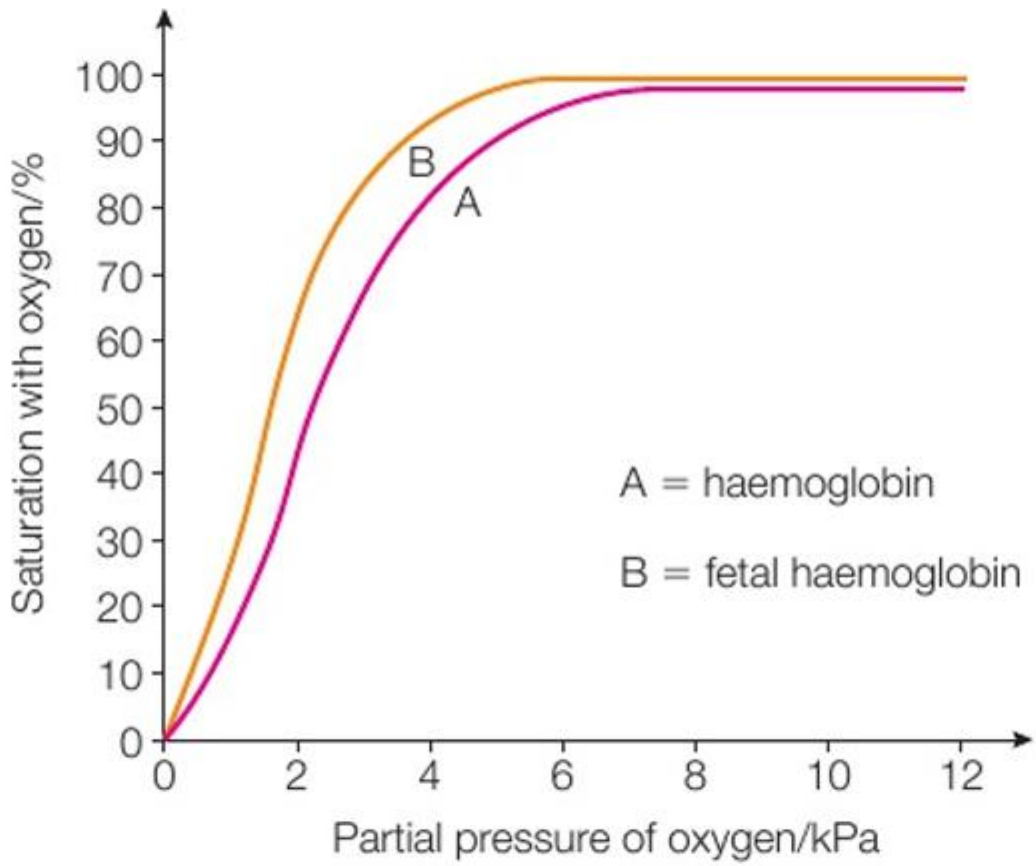


CO_2 Level	Curve Shift	Affinity	Effect
High CO_2	Right	↓ Decreases	Promotes O_2 unloading in tissues
Low CO_2	Left	↑ Increases	Promotes O_2 loading in lungs

Fetal Haemoglobin (HbF):

- **Fetal haemoglobin** has a **higher affinity** for O_2 than adult Hb.

- So the curve is **shifted** to the **left**
- Enables **O_2 transfer** from mother's blood across the placenta.
- At the same partial pressure of oxygen, affinity of fetal haemoglobin to oxygen is greater than that of the mother's adult haemoglobin.
- So, oxygen is released from adult haemoglobin and binds to fetal haemoglobin, so that the fetus receives oxygen for aerobic respiration



Myoglobin (not in specification but helps in some answers)

- Much higher affinity for oxygen than haemoglobin
- Releases oxygen at very low partial pressure
- Present in skeletal muscle cells so that oxygen is released during exercise, when rate of respiration is very high, large amounts of oxygen are needed and depleted to very low levels in short time
- So energy for exercise is released from ATP produced in aerobic respiration

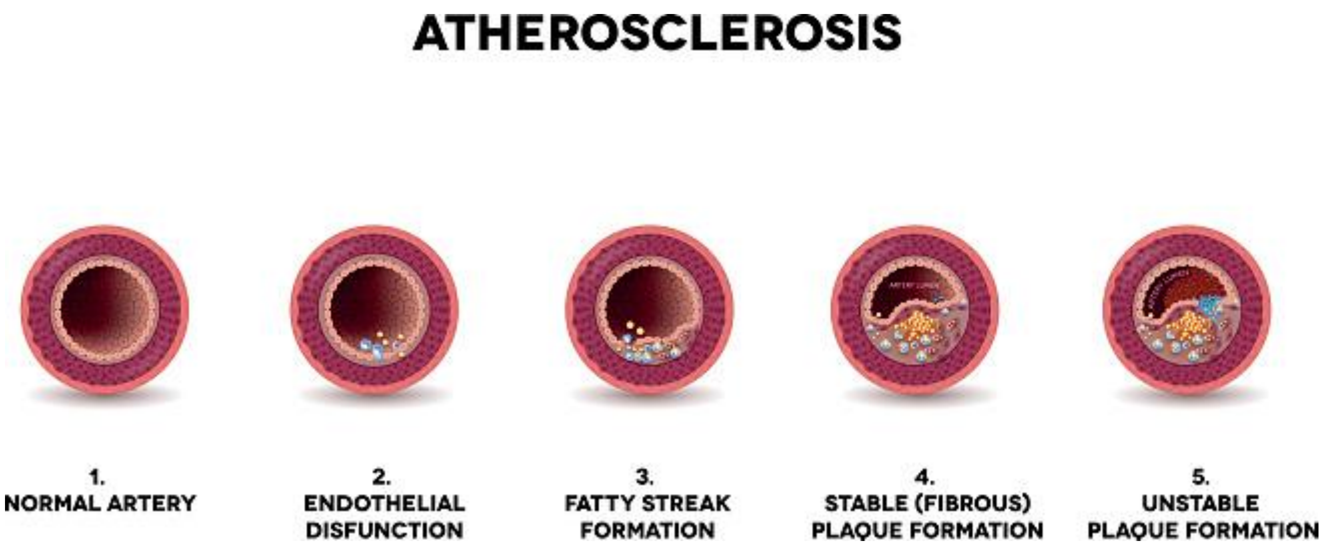
1.11 Atherosclerosis

A **progressive disease** that leads to narrowing and hardening of arteries due to plaque build-up.

Steps of Development:

1. **Endothelial damage**: from high BP, smoking, toxins.
2. **Inflammatory response**: white blood cells move to area.
3. **Fatty deposits (atheroma)**: cholesterol accumulates as well as calcium ions.
4. **Plaque formation**: fibrous cap forms over the atheroma.
5. **Narrowing of artery**: reduced lumen size, decreased blood flow.
6. **Increased BP**: due to resistance in narrowed vessels — positive feedback.

Leads to angina, myocardial infarction (heart attack), stroke, etc.



1.12 Blood Clotting and its Role in Cardiovascular Disease (CVD)

Essential to prevent excessive bleeding, but in arteries it can cause blockages leading to heart attacks and strokes.

Steps in Clotting Cascade:

1. **Platelet activation** and adherence to damaged endothelium.
2. **Thromboplastin release**: catalyzes prothrombin + Ca^{2+} → thrombin (enzyme).
3. **Thrombin**: converts **fibrinogen (soluble)** into **fibrin (insoluble)**.
4. **Fibrin mesh**: traps RBCs and platelets to form a stable clot.

Link to CVD:

- In arteries narrowed by atherosclerosis, clotting can lead to **complete blockage**, causing:
  - Myocardial infarction (heart attack)=> cut off oxygen and glucose delivery to heart cells and heart muscle cells, so heart muscles cannot contract
  - Cerebral infarction (stroke)
  - Aneurysm—localised dilation of artery in brain, may lead to rupture

1.13 Dietary Antioxidants and Cardiovascular Disease (CVD)

Free radicals and oxidative stress:

- **Free radicals** are unstable molecules with unpaired electrons.
- Formed during normal metabolism, smoking, pollution, etc.
- Can damage lipids, proteins, DNA — especially **endothelial cells** of arteries, initiating **atherosclerosis**.

Antioxidants:

- Molecules that **donate electrons** to free radicals, neutralizing them.
- Prevent **oxidation of LDL cholesterol** — a key event in plaque formation.

Examples:

- **Vitamin C**, **Vitamin E**, **β-carotene**, **flavonoids** (from fruits/veggies).
- Found in citrus fruits, green tea, berries, dark chocolate.

Link to CVD:

- **High antioxidant intake** may reduce risk of atherosclerosis and hence CVD.
- **BUT**: correlation ≠ causation — conflicting evidence from clinical trials.

1.14 – Core Practical: Vitamin C Content Investigation

1.15 Interpreting Quantitative Data on Illness and Mortality: Correlation vs Causation

Understanding Data:

- **Morbidity**: frequency of disease in a population.
- **Mortality**: death rate from a disease (e.g., CVD).
- Data is used to identify risk factors and population-level trends.

Correlation:

- A relationship between two variables (e.g., high-fat diet and CVD).
- **Does not prove** that one causes the other.

Causation:

- Proven through **controlled studies**, with:
  - Elimination of confounding variables.
  - Repeated, consistent results.

    - Known biological mechanism.

Conflicting Evidence:

- Studies can yield **different conclusions** due to:
    - Different sample sizes.
    - Biased reporting.
    - Variations in population and genetics.
    - Lack of control over confounders (e.g., activity levels, stress).

1.16 Evaluating Study Design: Validity and Reliability

To determine if a study gives trustworthy results, consider:

Sample selection:

- **Random** samples reduce selection bias.
- Must be **representative** of population (age, gender, ethnicity, etc.).

Sample size:

- **Larger** samples reduce the impact of anomalies and increase **statistical power**.
- Small samples = less reliable.

Control of variables:

- Must account for other factors (e.g., exercise, smoking) that could influence results.

Double-blind design(unit 2 aspect):

- Neither participants nor researchers know who is receiving the intervention.
- Eliminates **bias** in observation or reporting.
- Placebo overcomes psychological effects of receiving a new drug so that conclusions are valid
- Valid comparison is made using statistical tests such as t-tests (to check for significance in differences between observations in placebo group and the people receiving the intervention

Reproducibility:

- Studies must be replicable by others under similar conditions.
- Same results and conclusion reached by other methods

Outcome measures:

- Should be **quantifiable** (e.g., blood pressure, LDL levels).
- Avoid vague or self-reported measures (e.g., “feeling healthier”).

Repeatability

- ❖ Using very large sample size, controlling conditions and ensuring same experimental conditions for all groups and individuals in sample

1.17 Perception of Risk vs Actual Risk

People often make decisions based on **perceived risk**, not actual statistics. This can lead to **underestimating** or **overestimating** danger.

Why people misjudge risk:

- **Familiarity**: Common things (like smoking) seem safer. For example, a family member who is a persistent smoker may die at an old age while other family member who is a non-smoker may die at a young age.
- **Dread**: Rare but dramatic events (like a heart attack) are feared if seen in family members or friends.
- **Control**: Risks we “choose” (e.g., smoking, diet) feel less dangerous than imposed ones.
- **Media influence**: Sensational headlines distort real probabilities.

Implication:

- Public health campaigns must consider emotional biases.
- Risk communication needs both **numbers and narratives** to be effective.

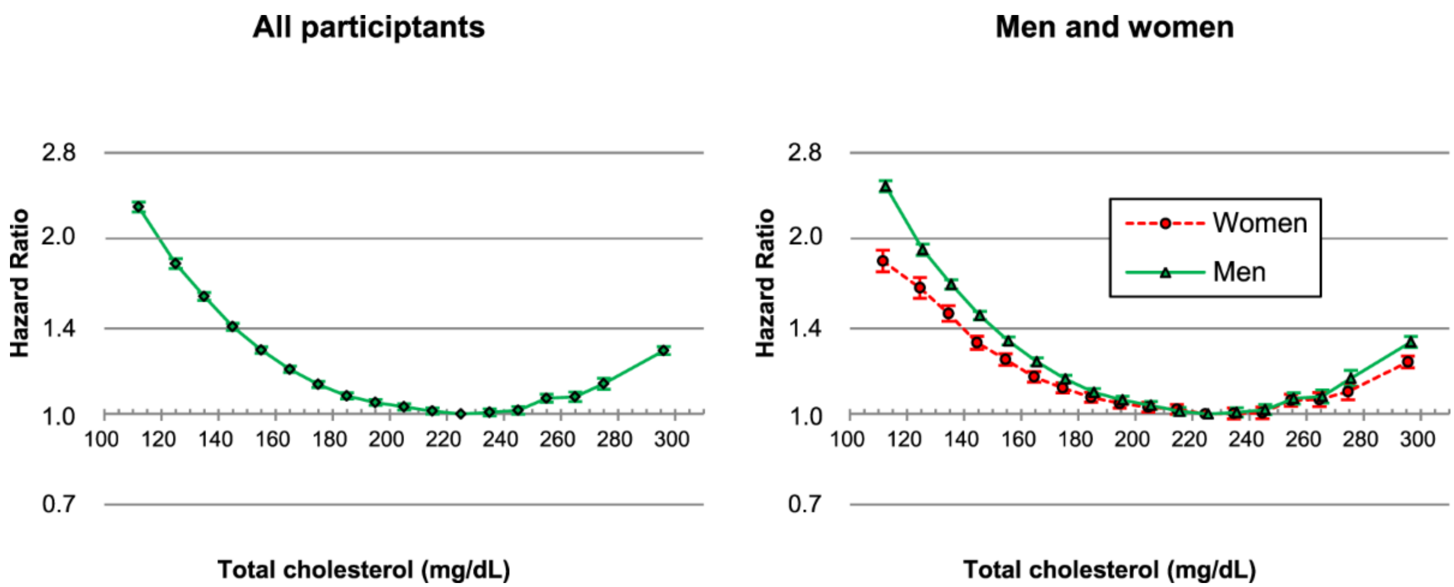
1.18 Cholesterol, HDLs, LDLs, and CVD

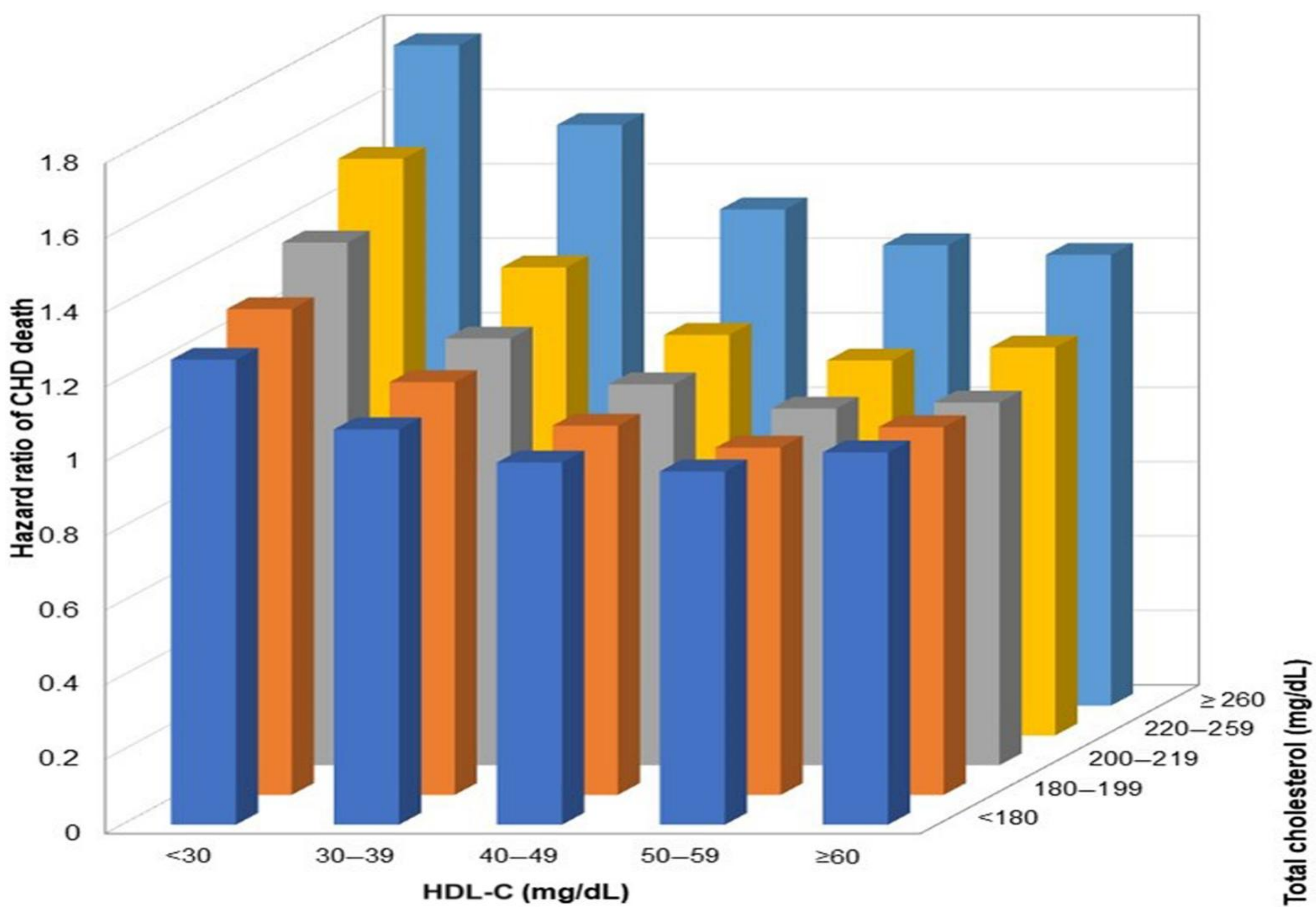
(i) Data Interpretation: HDL vs LDL:

- **LDLs (Low-Density Lipoproteins)**: transport cholesterol to tissues, including artery walls → promotes plaque formation.
- **HDLs (High-Density Lipoproteins)**: remove excess cholesterol from blood → transport to liver for excretion.

Blood Cholesterol Testing:

- Measures **total cholesterol**, HDL, LDL, and triglycerides.
- **Ratio of LDL:HDL** is key indicator — high LDL = increased CVD risk.





You may need to analyse similar graphs like this for exam.

Example: "Evaluate the data for CVD and risk of death"

Answer: The data demonstrates a non-linear (U-shaped) relationship between total cholesterol levels and the hazard ratio of CVD death. Risk is lowest around 200–220 mg/dL, increasing both below and above this range. This aligns with biological understanding: high total cholesterol increases CVD risk by promoting LDL-driven plaque formation, while very low cholesterol may impair essential cellular functions such as steroid hormone synthesis and membrane stability. The bar chart further shows that higher HDL-C levels (e.g. ≥60 mg/dL) consistently correlate with lower hazard ratios across all total cholesterol ranges, supporting HDL's role in reverse cholesterol transport and atheroma prevention.

However, while the data appears biologically valid, its application in accurately determining CVD risk is significantly limited. Most critically, there are **no error bars** on the bar chart or the "All Participants" line graph. This omission prevents evaluation of **data variability** or **statistical significance** between categories, making the reliability of observed trends uncertain. The line graph comparing men and women does show some error bars, but these are not quantified and do not overlap suggesting significant difference.

Furthermore, the **sample size is not stated**, and there is no information about the **age range, ethnicity, lifestyle factors like diet, smoking, activity, or health conditions** of the participants. These are important **confounding variables** that heavily influence CVD risk but are entirely uncontrolled in the dataset. For instance, individuals with high HDL-C may also be more physically active or have healthier diets, contributing independently to lower risk. Without stratifying the data or adjusting for these variables, conclusions about causation cannot be made — only correlations are shown.

The **hazard ratio** is also a **relative measure**, meaning it compares risk between groups but does not reflect the **absolute probability** of death. This limits its clinical usefulness, as two individuals with the same ratio may still have different actual risks depending on background health factors.

Another weakness is the **grouping of cholesterol and HDL-C into broad bands** in the bar chart, which may **oversimplify trends**. Sharp changes within a group could be hidden, and subtle interactions between variables may be lost. Moreover, there's no indication of whether cholesterol levels were **monitored over time** or measured only once — a key issue, since CVD is a chronic condition, and single-point measurements may not reflect long-term exposure or variability.

Lastly, the data does not specify the **time scale over which deaths were recorded**. Long-term studies are more reliable for assessing chronic disease outcomes. Without knowing the **follow-up duration**, it is difficult to assess whether observed deaths were directly linked to baseline cholesterol levels or due to unrelated acute events.

(ii) Evidence for Causal Relationship:

- High LDL levels **directly correlate** with increased plaque formation.
- Intervention trials (e.g., with statins) show **lowering LDL reduces CVD**.
- Biological mechanism is well understood — **causal** relationship is established, not just correlation.

1.19 Using Scientific Knowledge to Reduce Risk of Coronary Heart Disease (CHD)

Understanding the **modifiable risk factors** of CVD allows individuals to make informed lifestyle choices. Scientific knowledge about these factors is applied through public health campaigns and personal strategies.

1. Diet:

- Reduce intake of **saturated fats** → lowers LDL cholesterol, so cholesterol is less likely to be accumulated in areas of endothelial inflammation (lower chances of plaque buildup)
- Increase **unsaturated fats** (e.g. olive oil, fish oil) → raises HDL levels, so reduced LDL (and cholesterol transport in blood)
- Consume **antioxidants** (vitamin C, E, flavonoids) → neutralize free radicals.
- Increase **dietary fiber** → lowers cholesterol absorption.
- Avoid drinking **alcohol** → prevents liver damage and liver is essential for excretion of excess lipids

2. Obesity indicators:

- **Body Mass Index (BMI)** = weight (kg) ÷ height^2 (m^2)
    - Healthy: 18.5 – 24.9
    - Overweight: 25 – 29.9
    - Obese: 30+
    - Limitations: does **not account for muscle mass or fat distribution**
- **Waist-to-hip ratio (WHR)** = waist circumference ÷ hip circumference
    - Strong predictor of **visceral fat** → linked to CVD
    - Ideal WHR:
        - Men: <0.90
        - Women: <0.85

    Weight loss can be induced by:

- ❖ Increased exercise, reduced sugar, fat and cholesterol intake
- ❖ Calorie deficit, greater energy usage than consumption

3. Exercise:

- Regular **aerobic exercise**:
    - Increases HDL
    - Reduces blood pressure
    - Improves insulin sensitivity
    - Helps weight control
    - Strengthens muscles including heart

  - Reduces heart rate which reduces risk of hypertension

4. Smoking:

- Major risk factor:
  - Increases blood pressure
  - Increases risk of blood clotting
  - Damages endothelium → initiates atherosclerosis

Other non-modifiable risk factors
1. Age

- Risk **increases with age**.
- Arteries naturally become **less elastic and more stiff** which increases risk of damage to endothelial lining so the risk of atherosclerosis and atheroma increases, and plaque accumulates over time.

2. Sex / Gender

- **Men** are at a **higher risk** than premenopausal women.
- This is partly due to the **effects of oestrogen** in women, which:
  - Enhances HDL levels
  - Reduces LDL oxidation
- Post-menopause, **female risk rises sharply** and may equal or surpass male risk.

3. Genetic Factors / Family History

- Linked to inherited traits such as:
  - High LDL cholesterol (e.g. familial hypercholesterolemia)
  - Hypertension predisposition
  - metabolism disorders

1.20 Treatments for Cardiovascular Disease (CVD): Benefits and Risks

Pharmacological interventions are prescribed based on individual risk profiles. Each treatment has associated benefits and risks which must be **critically assessed**.

1. Antihypertensives

Used to lower **high blood pressure**, a key CVD risk.

- **Types**:
  - ACE inhibitors: inhibit vasoconstriction.
  - Beta blockers: reduce heart rate and output.

- Calcium channel blockers: reduces blood pressure by reducing heartbeats

- **Benefits**:
  - Reduces endothelial damage and risk of stroke/heart attack.
- **Risks**:
  - Dizziness, fainting (due to low BP)
  - Kidney dysfunction
  - Depression or cold extremities (especially with beta blockers)

2. Statins

Lower **blood cholesterol**, especially LDL.

- **Mechanism**:
  - Inhibit HMG-CoA reductase enzyme in cholesterol synthesis.
- **Benefits**:
  - Significant reduction in CVD events.
  - Can be used **preventively** in high-risk individuals.
- **Risks**:
  - Muscle pain and weakness
  - Liver damage
  - Rarely: memory loss or increased diabetes risk
  - Stomachache
  - Diarrhoea
  - Sleeping problems and disorders

3. Anticoagulants

Prevent **clot formation**.

- **Common drugs**: warfarin, heparin.
- **Benefits**:
  - Reduces risk of thrombus (clot) forming in narrowed arteries.
  - Prevents embolism → protects against heart attack and stroke.
- **Risks**:
  - Increased bleeding risk (even from minor injuries)

- Requires careful **dosage monitoring** (INR for warfarin)
- Drug interactions common
- Internal bleeding
- Uncontrollable bleeding

4. Platelet Inhibitors

Prevent platelet aggregation.

- **Example**: aspirin (irreversibly inhibits COX enzyme).
- **Benefits**:
    - Reduces risk of clot formation at plaques.
    - Often used in combination with other drugs.
- **Risks**:
    - Stomach ulcers or gastrointestinal bleeding.
    - Allergic reactions.

Evaluating Treatments:

- Choice depends on:
    - **Risk-benefit ratio** for the patient
    - Presence of **multiple conditions**
    - Cost and accessibility
    - Patient compliance and **side-effect tolerance**
    - Doctor's advice (after checking patient profile, measures such as genetic screening etc)

Topic 2: Membranes, Proteins, DNA and Gene Expression

2.1 Properties of Gas Exchange Surfaces + Fick's Law

Essential Properties of Efficient Gas Exchange Surfaces:

1. **Large surface area**
    - Increases space for diffusion
    - Example: alveoli in lungs (≈300 million)
2. **Thin barrier (short diffusion distance)**
    - Speeds up diffusion